

PHYSICS PRACTICAL SHEETS

....KV.....-Campus

Date 4 June 2026Class : CE-1/11Roll No.: 43Shift: Morning

Object of the Experiment (Block Letter)

*Rausul Faruqi*Experiment No.: 1Group : SSub: PHYS 102 LabSet: 1

MEASUREMENT OF LOW RESISTANCE USING CAREY-FOSTER'S BRIDGE.

Apparatus required:

- 1) Carey-Foster's bridge
- 2) Two equal resistances
- 3) A fractional resistance box
- 4) Cell
- 5) Connecting wires
- 6) A sensitive galvanometer and a jockey

Theory:

The Carey-Foster Bridge is an advanced version of the meter bridge. It is designed for precise measurements and is commonly used to determine small changes in resistance. Its arrangement is similar to that of a Wheatstone bridge, providing high accuracy.

The unknown resistance, X , is determined using the equation

$$X = Y - Sd \quad \text{--- (1)}$$

where Y is a known resistance, S represents the resistance per unit length of the Carey Foster bridge wire and d is the corrected difference in balance point readings, calculated as:

$$d = S - \delta L \quad \text{--- (2)}$$

where S is the observed difference in balance point readings when the positions of resistances X and Y are interchanged and δL is the applied correction factor.

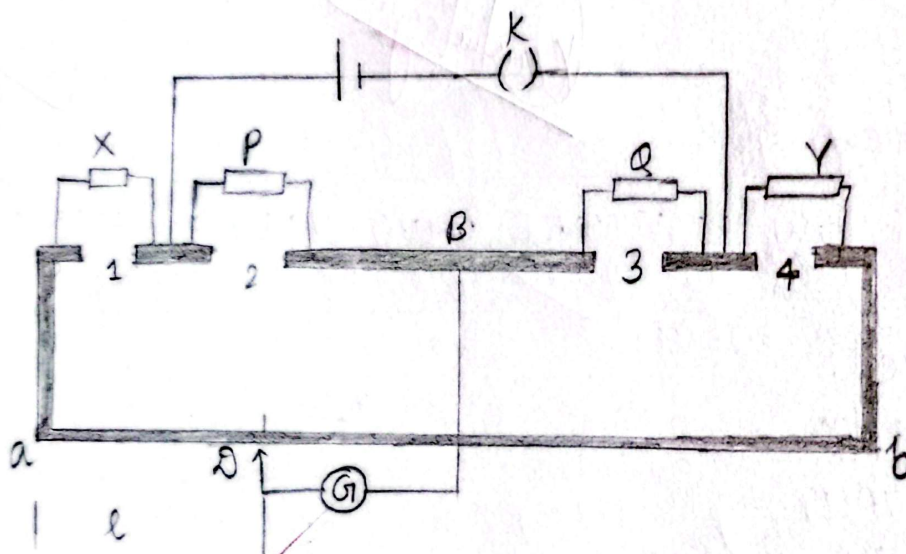


Fig 1: Schematic diagram of the circuit of Carey-Foster's bridge.
 P & Q are equal resistances.

Observations

Balance point with Y zero in the right gap and X zero in the left gap, $x = 53.5 \text{ cm}$

Balance point with Y zero in the left gap and X zero in the right gap, $y = 38 \text{ cm}$

Correction to be applied, $\delta L = (x - y) = 15.5 \text{ cm}$

Table 1: Table to determine resistance per centimeter of Carey Foster wire

No. of obs	X Ω	Y Ω	Position of the balance point with copper strip in		Shift $s = (l_1 - l_2)$ cm	Corrected shift $d = s - \delta L$ cm	Resistance per cm $S = \frac{Y}{d} \frac{\Omega}{\text{cm}}$
			Left gap l_1 cm	Right gap l_2 cm			
1	0	0.1	54	34	20	4.5	0.02
2	0	0.2	59.5	39.5	20	4.5	0.04
3	0	0.3	65.5	27.5	38	22.5	0.01
4	0	0.4	60.5	33	27.5	12	0.03
5	0	0.5	62	19.5	42.5	27	0.02

Mean $S = 0.02 \Omega/\text{cm}$

The resistance per unit length of the Carey Foster bridge wire is 0.02.

Table 2: Table for determining unknown resistance

No. of obs	Y Ω	Position of the balance point with the unknown resistance in		Shift $S = (l_1 - l_2)$ cm	Corrected shift $d = S - dL$ cm	Unknown resistance $X = Y - Sd \Omega$
		Left gap l_1 , cm	Right gap l_2 , cm			
1	0.1	54.5	38	16.5	1	0.08
2	0.2	55.5	34.5	21	5.5	0.09
3	0.3	53.5	37.5	16	1.5	0.07
4	0.4	57	39.5	17.5	2	0.36
5	0.5	55	37.5	17.5	2	0.46

Mean $X = 0.212 \Omega$

Result: The unknown low resistance was determined to be 0.212Ω using the Carey-Foster Bridge.

Precautions:

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17th June 2026

- 1) Use thick copper leads to connect the unknown resistance which helps reduce additional resistance in the circuit.
- 2) Confirm that all plugs in the resistance box are firmly seated to maintain consistent and accurate resistance values.
- 3) Handle the jockey wire care — apply it tightly and avoid applying pressure or dragging it along the wire to prevent damage and maintain precision in measurement.

Conclusion: The resistance per unit length and unknown low resistance was determined.